

# The Maximum Subarray

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We define subsequence as any subset of an array. We define a subarray as a contiguous subsequence in an array.

Given an array, find the maximum possible sum among:

1. all nonempty subarrays.
2. all nonempty subsequences.

Print the two values as space-separated integers on one line.

**Note** that empty subarrays/subsequences should not be considered.

## Example

`arr = [-1, 2, 3, -4, 5, 10]`

The maximum subarray sum is comprised of elements at indices `[1 - 5]`. Their sum is  $2 + 3 + -4 + 5 + 10 = 16$ . The maximum subsequence sum is comprised of elements at indices `[1, 2, 4, 5]` and their sum is  $2 + 3 + 5 + 10 = 20$ .

## Function Description

Complete the `maxSubarray` function in the editor below.

`maxSubarray` has the following parameter(s):

- `int arr[n]`: an array of integers

## Returns

- `int[2]`: the maximum subarray and subsequence sums

## Input Format

The first line of input contains a single integer `t`, the number of test cases.

The first line of each test case contains a single integer `n`.

The second line contains `n` space-separated integers `arr[i]` where  $0 \leq i < n$ .

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Difficulty: **Medium**  
Max Score: 50  
Submitted By: 84131

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```
1 import java.io.*;
2 import java.util.*;
3 import java.util.stream.*;
4 import static java.util.stream.Collectors.joining;
5 import static java.util.stream.Collectors.toList;
6
7 class Result {
8
9     /*
10      * Complete the 'maxSubarray' function below.
11      *
12      * The function is expected to return an INTEGER_ARRAY.
13      * The function accepts INTEGER_ARRAY arr as parameter.
14      */
15
16     public static List<Integer> maxSubarray(List<Integer> arr) {
17
18         // Kadane's algorithm for maximum subarray
19         int currentSum = arr.get(0);
20         int maxSubarraySum = arr.get(0);
21
22         // Maximum non-contiguous sum
23         int maxSubsequenceSum = arr.get(0);
24
25         for (int i = 1; i < arr.size(); i++) {
26
27             int value = arr.get(i);
28
29             // Maximum contiguous subarray
30             currentSum = Math.max(value, currentSum + value);
31             maxSubarraySum = Math.max(maxSubarraySum, currentSum);
32
33             // Maximum non-contiguous subsequence
34             if (value > 0) {
35                 maxSubsequenceSum += value;
```



35

maxSubsequenceSum += value;

Line: 93 Col: 2

Upload Code as File



Test against custom input

Run Code

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## Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Sample Test case 1

Input (stdin)

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```
1 2
2 4
3 1 2 3 4
4 6
5 2 -1 2 3 4 -5
```

Your Output (stdout)

```
1 10 10
2 10 11
```

Expected Output

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